CHAPTER 8

ALDEHYDES, KETONES AND CARBOXYLIC ACIDS

CHAPTER ESSENCE AT A GLANCE:
Nucleophilic addition across the carbonyl group, aldol condensation, Cannizzaro reaction, oxidation-reduction tests, and carboxylic acid acidity.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Carbonyl Polar Structure & Addition:** The polar carbon-oxygen double bond imparts strong electrophilic character to the carbonyl carbon, driving nucleophilic additions of cyanide, bisulphite, and ammonia derivatives.

• **Aldol Condensation & Cannizzaro Disproportionation:** Aldehydes with alpha-hydrogens undergo base-catalyzed aldol condensation to give beta-hydroxy carbonyls that dehydrate into alpha-beta unsaturated products; aldehydes lacking alpha-hydrogens undergo Cannizzaro redox disproportionation.

• **Distinguishing Oxidation & Reduction Tests:** Aldehydes are differentiated from ketones through selective oxidation by Tollens' silver mirror and Fehling's copper reduction tests, while methyl carbonyls yield yellow iodoform precipitates.

• **Carboxylic Acid Acidity & Resonance:** Carboxylic acids display pronounced acidity due to resonance stabilization of the carboxylate anion across two equivalent oxygen atoms; electron-withdrawing substituents increase acid strength by dispersing negative charge.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$\text{R-COCl} + \text{H}_2 \xrightarrow{\text{Pd / BaSO}_4\text{, quinoline}} \text{R-CHO} + \text{HCl} \quad \text{(Rosenmund)}$$

Rosenmund reduction of acyl chlorides to corresponding aldehydes

$$2\text{CH}_3\text{CHO} \xrightarrow{\text{dil. NaOH}} \text{CH}_3\text{-CH(OH)-CH}_2\text{-CHO} \xrightarrow{\Delta} \text{CH}_3\text{-CH=CH-CHO}$$

Aldol condensation and subsequent dehydration to alpha-beta unsaturated aldehyde

$$2\text{HCHO} + \text{conc. KOH} \rightarrow \text{CH}_3\text{OH} + \text{HCOOK} \quad \text{(Cannizzaro)}$$

Cannizzaro disproportionation of non-enolizable aldehydes

$$> \text{C=O} \xrightarrow{\text{Zn-Hg / conc. HCl}} > \text{CH}_2 \quad \text{(Clemmensen Reduction)}$$

Clemmensen reduction of carbonyl groups directly to methylene hydrocarbons

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Aldol requires PRESENCE of α-hydrogen; Cannizzaro requires ABSENCE of α-hydrogen!

• **Alert:** [Valuation Rubric] Identify whether aldehyde possesses α-hydrogen atoms .

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Explain the preparation of aldehydes by: (a) Rosenmund reduction, (b) Stephen reduction, (c) Etard reaction. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Rosenmund Reduction]** Acyl chloride hydrogenated over Pd catalyst supported on $BaSO_4$ poisoned with sulfur/quinoline: R-COCl + H_2 → [Pd divided by $BaSO_4$] R-CHO + HCl. [1 Mark]

• **[Step 2: Stephen Reaction]** Nitriles reduced with $SnCl_2$ and HCl: R-CN + $SnCl_2$ + 2HCl --> R-CH=NH·HCl --> [H_3O^+] R-CHO + NH_4Cl. [1 Mark]

• **[Step 3: Etard Reaction]** Toluene oxidized with chromyl chloride: C_6H_5-CH_3 + 2CrO_2Cl_2 (in CS_2) --> chromium complex --> [H_3O^+] C_6H_5-CHO (benzaldehyde). [1 Mark]

• **Key Exam Result:** Rosenmund (Pd divided by $BaSO_4$), Stephen ($SnCl_2$ divided by HCl), Etard (CrO_2Cl_2)

2. Explain the mechanism of nucleophilic addition of HCN and sodium bisulphite (NaHSO3) to carbonyl compounds. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Carbonyl Geometry]** Carbonyl carbon is sp^2 hybridized and planar. Nucleophile attacks sp^2 carbon perpendicularly (at ~107° angle), forming tetrahedral alkoxide intermediate. [1.5 Marks]

• **[Step 2: Addition of HCN & $NaHSO_3$]** HCN: $:CN^-$ attacks >C=O to form cyanohydrin >C(OH)(CN). Base catalyst generates $:CN^-$. $NaHSO_3$: HSO_3^- attacks to form bisulphite addition product >C(OH)($SO_3^-Na^+$) (crystalline solid). [1.5 Marks]

• **Key Exam Result:** Nucleophilic addition forming tetrahedral alkoxide intermediate

3. Explain Aldol condensation and Cross-Aldol condensation with balanced chemical equations and conditions. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Aldol Condensation]** Two molecules of aldehyde/ketone having α-hydrogen react in presence of dilute NaOH: 2 CH_3CHO → [dil. NaOH] CH_3-CH(OH)-CH_2-CHO (3-hydroxybutanal / aldol) → [Heat, -H_2O] CH_3-CH=CH-CHO (but-2-enal). [2 Marks]

• **[Step 2: Cross-Aldol Condensation]** Between two different aldehydes (e.g. ethanal and propanal); yields a mixture of 4 products (two self-aldol and two cross-aldol products). [2 Marks]

• **Key Exam Result:** Requires α-H; forms β-hydroxy aldehyde which dehydrates to α,β-unsaturated aldehyde

4. Explain Cannizzaro reaction and Cross-Cannizzaro reaction with equations. Which aldehydes undergo this reaction? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Cannizzaro Reaction]** Aldehydes with NO α-hydrogen undergo disproportionation (self-redox) with concentrated alkali: 2 HCHO + conc. NaOH --> CH_3OH (reduced) + HCOONa (oxidized). 2 C_6H_5CHO + conc. KOH --> $C_6H_5CH_2OH$ (benzyl alcohol) + C_6H_5COOK (potassium benzoate). [2 Marks]

• **[Step 2: Cross-Cannizzaro]** HCHO + C_6H_5CHO + conc. NaOH --> HCOONa + $C_6H_5CH_2OH$ (formaldehyde is more easily oxidized). [1 Mark]

• **Key Exam Result:** Aldehydes without α-H undergo redox: 1 alcohol + 1 carboxylate salt

5. Explain reduction of aldehydes and ketones by: (a) Clemmensen reduction, (b) Wolff-Kishner reduction. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Clemmensen Reduction]** Carbonyl group reduced to -CH_2- using zinc amalgam and concentrated HCl: >C=O + 4[H] → [Zn-Hg / conc. HCl] -CH_2- + H_2O. (e.g. CH_3COCH_3 → $CH_3CH_2CH_3$). [1.5 Marks]

• **[Step 2: Wolff-Kishner Reduction]** Carbonyl group heated with hydrazine followed by KOH in ethylene glycol (453-473 K): >C=O + NH_2NH_2 → >C=NNH_2 + H_2O → [KOH / glycol, heat] -CH_2- + $N_2\uparrow$. [1.5 Marks]

• **Key Exam Result:** Clemmensen (Zn-Hg / conc. HCl) | Wolff-Kishner (NH_2NH_2 divided by KOH)

6. Explain Hell-Volhard-Zelinsky (HVZ) reaction for alpha-halogenation of carboxylic acids with equation. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Reaction Statement]** Carboxylic acids having α-hydrogen react with chlorine or bromine in presence of small amount of red phosphorus followed by hydrolysis to form α-halocarboxylic acids: $R\text{-}CH_2\text{-}COOH + X_2$ --> [Red P divided by H_2O] $R\text{-}CH(X)\text{-}COOH + HX$ (X = Cl, Br). [2 Marks]

• **Key Exam Result:** HVZ Reaction: α-halogenation with X_2 divided by Red Phosphorus

7. Explain the decarboxylation of carboxylic acids using soda lime with balanced equation. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Decarboxylation]** Sodium salts of carboxylic acids heated with soda lime (mixture of NaOH and CaO in 3:1 ratio): R-COONa + NaOH --> [CaO, heat] R-H (alkane with one less carbon) + Na_2CO_3. e.g. $CH_3COONa + NaOH \rightarrow CH_4 + Na_2CO_3$. CaO keeps NaOH dry and prevents glass fusion. [2 Marks]

• **Key Exam Result:** R-COONa + NaOH or CaO --> R-H + Na_2CO_3 (decarboxylation)